# A.L.T.R.U.S – Adaptive Life-support & Therapeutic Robotic Unit System

Project ID: 25-26j-437

Project Proposal

Madushanka L.A.S – IT22602114

Supervised by: Ms. Hansi De Silva

Co-supervised by: Mr. Ashvinda Iddamalgoda

B.Sc. (Hons) Degree in Information Technology

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

# DECLARATION

We are confirming that the work given in this proposal is one of our own volition. The same work has not been given, entirely or partially, for academic credit at any other university or institution. We have also made sure that all sources of information and ideas from others have been clearly cited and appropriately acknowledged in the document.

| Name | Student ID | Signature |
|---|---|---|
| Madushanka L.A.S | IT22602114 | |

The above candidates are researching the undergraduate Dissertation under my supervision.

Name of supervisor: Ms. Hansi De Silva

Name of co-supervisor: Mr. Ashvinda Iddamalgoda

Signature of supervisor: Date: 2025/08/29

Signature of co-supervisor: Date: 2025/08/29

# ABSTRACT

This article presents A.L.T.R.U.S (Adaptive Life-support & Therapeutic Robotic Unit System), a context-aware, modular assistive robot designed for safe, smart, and empathetic behavior in dynamic home scenarios. The system integrates four core software frameworks within a ROS2-based architecture to overcome the majority of shortcomings of current assistive robotics. An adaptive navigation module initially combines static SLAM with dynamic real-time LIDAR-based updates according to RSO2 logic to enable dynamic replanning of paths in response to environmental change—remapping is not needed. Voice and hand gesture control is integrated directly into the navigation layer so that users can call, pause, or drive the robot by natural voice commands or by a hand wave, enabling much enhanced accessibility for the disabled and elderly. Second, an environmental and health monitoring system in real-time combines data from environmental sensors (MQ-2, DHT11) and vital sign sensors (MPU6050, MAX30100) to detect emergencies like gas leaks, fever, or falls and provide automatic alerts via Firebase and Twilio. Third, an emotion-perceiving HRI module detects facial expressions and voice tone with OpenCV and pre-trained models (RAVDESS, FER2013), providing empathetic responses through a chatbot based on Dialogflow. Fourth, a fault-tolerant middleware offers system robustness through auto-recovery, intent arbitration, plugin update security, and blockchain log-based recording with Hyperledger Fabric for trust and transparency. Based on a modular ROS2 foundation, A.L.T.R.U.S brings together adaptive navigation, proactive healthcare, emotional intelligence, and robust software infrastructure in a unified, edge-deployable platform. This student innovation stretches the limits of assistive robotics by combining autonomy, safety, empathy, and security, which is ideal for smart homes, elderly care, and independent living support.

**Keywords:**

**Adaptive Navigation, Dynamic SLAM, Human-Robot Interaction, Voice Command, Gesture Recognition, Emotion Recognition, Assistive Robotics, ROS2, Sensor Fusion, Blockchain, Proactive Healthcare**

# TABLE OF CONTENTS

DECLARATION ..... 2
ABSTRACT ..... 3
TABLE OF CONTENTS ..... 4
LIST OF FIGURES ..... 6
LIST OF TABLES ..... 7
LIST OF ABBREVIATIONS ..... 8
1. INTRODUCTION ..... 9
1.1 BACKGROUND ..... 9
1.2 LITERATURE SURVEY ..... 12
1.2.1 Facial Expression Recognition in Assistive Robotics ..... 12
1.2.2 Vocal Emotion Recognition for Social and Assistive Robots ..... 12
1.2.3 Multimodal Emotion Recognition Systems ..... 12
1.2.4 Shortcomings of Existing Literature ..... 13
1.3 RESEARCH GAP ..... 14
1.4 RESEARCH PROBLEM ..... 16
2. RESEARCH OBJECTIVES ..... 17
2.1 MAIN OBJECTIVE ..... 17
2.2 SPECIFIC OBJECTIVES ..... 17
3. METHODOLOGY ..... 19
3.1 Multimodal Emotion and Command Detection ..... 19
3.1.1 Facial Emotion Recognition (FER) ..... 19
3.1.2 Vocal Tone Analysis ..... 19
3.2 Multimodal Fusion and Context-Aware Decision Making ..... 19
3.3 Empathetic Speech Response Generation ..... 20
3.4 System Architecture Overview ..... 20
3.5 Project Flow ..... 20
3.6 Evaluation Plan ..... 20
3.7 Commercialization ..... 21
3.8 Overall System Diagram ..... 22

4. PROJECT REQUIREMENTS ..... 23
4.1 Functional Requirements ..... 23
4.2 Non-Functional Requirements ..... 24
5. BUDGET DETAILS ..... 25
6. CONCLUSION ..... 26
REFERENCES ..... 27
References ..... 27
APPENDICES ..... 28
Appendix 1. Gantt Chart ..... 28
Appendix 2. Online Survey ..... 29
Appendix 3. Work Breakdown Structure ..... 32

# LIST OF FIGURES

Figure 1.1.1 – Survey results on prior interaction with robots or virtual assistants ..... 10
Figure 2.1.2 – Survey results on comfort levels when interacting with robots ..... 10
Figure 3.1.3 – Survey results on the most important emotions for robots to detect ..... 11
Figure 4.3.1 System Overview Diagram ..... 22

# LIST OF TABLES

Table 1 budget for the proposed system ..... 25

## LIST OF ABBREVIATIONS

| ABBREVIATION | DESCRIPTION |
|---|---|
| HRI | Human-Robot Interaction |
| AI | Artificial Intelligence |
| FER | Facial Emotion Recognition |
| CNN | Convolutional Neural Network |
| ML | Machine Learning |
| DL | Deep Learning |
| CV | Computer Vision |
| NLP | Natural Language Processing |

# 1. INTRODUCTION

## 1.1 BACKGROUND

The integration of robotics into daily human life is rapidly shifting from industrial automation to personal assistance, healthcare support, and education. As these roles become more human-facing, a critical challenge emerges. Robots must not only perform tasks efficiently but also communicate in ways that feel natural, empathetic, and human-like. While current assistive robots can process instructions, navigate environments, and execute complex actions, most remain emotionally unaware, treating human interactions as transactional rather than relational. This emotional gap limits their acceptance, trustworthiness, and long-term utility in sensitive environments such as elder care, therapy, and education.

This research focuses on bridging that gap through the development of a real-time, multi-modal emotion-aware interaction model tailored for assistive robots. The system will integrate facial expression recognition and vocal tone analysis into a unified emotional understanding engine. For the initial phase, the scope will be narrowed to happiness and sadness detection. Two distinct emotional extremes that are relatively easier to distinguish, cover both positive and negative affect, and form a foundation for scaling to more complex emotions in the future.

The project's novelty lies in its multi-modal data fusion and low-latency processing design. Unlike existing systems that rely on a single input channel, the proposed model will simultaneously analyze visual and auditory signals, cross-validating results to minimize errors. This enables the robot to respond almost instantly with empathetic speech, enhancing human-robot trust and rapport. By grounding development in both cutting-edge AI techniques and real-world emotional needs, this research moves toward a future where assistive robots are not just intelligent, but socially aware and emotionally adaptive.

User Feedback to Support the Research

A survey was conducted with 20+ participants to gather public perspectives on emotion-aware robots. The results provided valuable insights into how people perceive and interact with such systems.

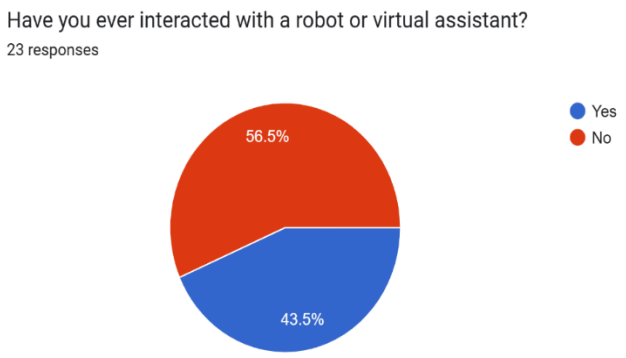



*Figure 1.1.1 – Survey results on prior interaction with robots or virtual assistants*

Figure 1.1.1 illustrates participants whether they had ever interacted with a robot or a virtual assistant. 43.5% of the respondents answered "Yes", indicating prior exposure to AI-driven or automated systems such as chatbots, voice assistant, or service robots. The remaining 56.5% had never engaged with such technology, highlighting that while these systems are becoming more common, there is still a significant portion of the population without first-hand experience. This mix of familiarity and unfamiliarity is important when considering how users might respond to new, emotion-aware interaction models.

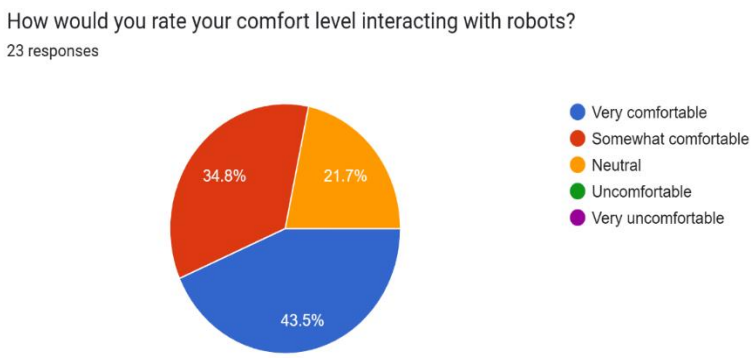



*Figure 2.1.2 – Survey results on comfort levels when interacting with robots*

Figure 1.1.2 measured participant's comfort levels when interacting with robots. Results showed that 43.5% felt very comfortable, 34.8% were somewhat comfortable, and 21.7% maintained a neutral stance. Interestingly, there were no responses for uncomfortable or very uncomfortable. This suggest that, in general, participants are either open to or at least accepting of robotic interaction, which creates a promising environment for introducing more advanced, emotionally intelligent systems.

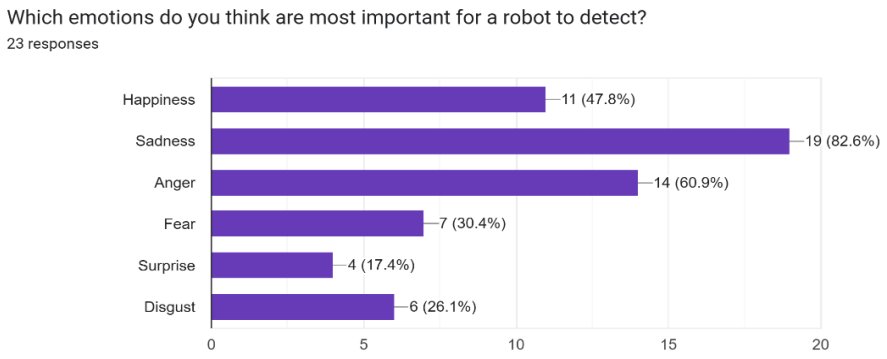



*Figure 3.1.3 – Survey results on the most important emotions for robots to detect*

Figure 1.1.3 explored which emotions participants considered most important for a robot to detect. Sadness ranked the highest (82.6%), followed by anger (60.9%) and happiness (47.8%). Other emotions such as fear, disgust and surprise were viewed as less essential. This suggests that people prioritize robots being able to respond empathetically to navigate or critical emotional states, likely because such recognition could lead to more supportive and helpful responses in real-life scenarios.

These findings confirm that there is a strong user demand for intelligent, responsive, and easy-to-use assistive robots that can adapt to changing environments and provide accessible multimodal control.

## 1.2 LITERATURE SURVEY

This section provides an overview of the latest research and technology in terms of emotion recognition and empathetic human-robot interaction for assistive robotics, and compares their strengths and limitations to the system proposed in this study.

### 1.2.1 Facial Expression Recognition in Assistive Robotics

Facial expression analysis remains one of the most widely adopted methods for emotion detection in assistive robots.

- **Deep Learning Approaches –** This [1]proposed a lightweight convolutional neural network for real-time facial expression recognition, achieving high classification accuracy on the FER2013 dataset while maintaining low computational cost, making it suitable for embedded systems. Similarly, [2]implemented a residual network architecture optimized for real-time robot perception. While these systems achieve strong accuracy, they often require high-quality imaging conditions and powerful hardware, limiting deployment on low-cost platforms.
- **Applications in Special Populations –** This [3]demonstrated the use of facial emotion detection in robots designed for children with Autism Spectrum Disorder (ASD), improving engagement during therapy sessions. However, the models still suffered from performance drops in varied lightning and occlusion-heavy environments, highlighting the need for robust pre-processing and data augmentation in real-time systems.

### 1.2.2 Vocal Emotion Recognition for Social and Assistive Robots

Speech tone and sentiment provide additional cues to user's emotional state, allowing robots to respond in a more human-like manner.

- **Acoustic Feature-Based Methods –** This [4]used Mel-Frequency Cepstral Coefficients (MFCCs) and spectrogram features combined with recurrent neural networks (RNNs) for emotion classification in speech.
- **NLP-Enhanced Speech Analysis –** This [5]combined acoustic emotion detection with Natural Language Processing (NLP) for context-sensitive empathetic responses. While effective in lab conditions, these systems degrade in home environments due to background noise and room acoustics unless advanced noise suppression techniques are employed.

### 1.2.3 Multimodal Emotion Recognition Systems

Combining multiple input modalities often yields better performance than single-source detection.

- **Facial + Voice Fusion –** This [6]developed a multimodal fusion model integrating facial expression and speech emotion recognition, achieving improved classification robustness in noisy conditions. Similarly, [7]designed a transformer-based architecture for fusing facial and audio features, outperforming unimodal systems in both accuracy and resilience to environmental variability. Pepper by SoftBank Robotics also implements facial and vocal emotion recognition to generate adaptive responses. However, these systems often target a wide range of emotions, increasing computational requirements and classification ambiguities.
- **Real-Time Constraints –** While multimodal fusion improves accuracy, it often comes at the cost of computational efficiency. Many existing implementations are designed for high-powered commercial robots and are unsuitable for resource-constrained assistive platforms.

### 1.2.4 Shortcomings of Existing Literature

Based on the analysis of existing studies, the following gaps are identified:

- Most emotion-aware systems attempt to detect a wide range of emotions, increasing complexity and reducing accuracy for real-time, low resource operation.
- Many facial and vocal recognition models are computationally heavy and require high-end hardware, limiting accessibility for low-cost assistive robots.
- Multimodal emotion recognition systems are often built for general-purpose social robots, not specifically for assistive care contexts.
- Empathetic response generation is usually scripted or predefined, lacking adaptive, context-aware interaction tailored to specific users.

The system presented in this research addresses these gaps by designing a lightweight, real-time multimodal emotion recognition framework that focuses on detecting only two key emotions (happy and sad), integrates both facial expression and vocal tone analysis, and produces contextually appropriate empathetic speech responses for assistive home environments.

## 1.3 RESEARCH GAP

Despite significant advances in HRI, affective computing, and multimodal perception, current emotion-aware assistive robots face several limitations when deployed in real-world home environments. Especially for elderly or physically impaired users who require simple, fast, and empathetic interaction.

- **Overgeneralized Emotion Sets –** Most emotion recognition systems in social or assistive robotics attempt to classify a broad range of 6-8 emotions (e.g., happy, sad, angry, etc.). While academically appealing, this approach increases computational complexity and classification errors, especially in noisy, uncontrolled home environments.
  Gap: No widely adopted assistive systems focuses on only the most context-relevant emotions (happy and sad) to improve accuracy and speed of low-resource embedded platforms.
- **Environment Sensitivity –** Existing models perform well in controlled lab conditions but suffer accuracy drops under varied lightning, occlusion, or background noise. For example, vocal models degrade with TV or fan noise, and facial models fail with low lighting or partial face visibility.
  Gap: Few systems employ robust pre-processing and noise suppression tailored for home environmental variability.
- **High Computational Demands –** State-of-the-art facial and speech emotion recognition models often rely on heavy convolutional or transformer architectures that require GPUs or high-end CPUs. This makes them unsuitable for low-cost, small-form-factor assistive robots.
  Gap: There is a lack of lightweight, real-time multimodal emotion recognition frameworks optimized for affordable, resource-constrained home robots.
- **Limited Empathetic Adaption –** Most emotion aware robots produce scripted responses rather than dynamically tailoring tone, pace, and content to the user's detected emotional state.
  Gap: There is a lack of context-aware empathetic speech generation that adapts both linguistically and prosodically to user mood.
- **Commercial vs. Assistive Context Misalignment –** Robots like Pepper and Ryan implement multimodal emotion recognition but are designed for broad social or entertainment purposes. They are expensive, overengineered for simple home-assistive use, and not optimized for real-time interaction in resource-limited platforms.
  Gap: There is no low-cost, purpose-built emotion-aware assistive robot that combines facial and vocal cues with adaptive empathetic responses for daily home care.

| Feature | Pepper | Ryan | HRI Research | Proposed System |
|---|---|---|---|---|
| Primary Purpose | Social interaction, public engagement | Elderly companionship, memory aids | Research on multimodal HRI & emotion detection | Assistive home care, focused on empathetic daily interaction |
| Emotion Detection Scope | 6-8 basic emotions | 6-8 basic emotions | 4-8 emotions | Only “Happy” and “Sad” for optimized accuracy and speed |
| Modalities Used | Facial recognition + speech tone + speech content | Facial recognition + speech tone | Varies | Facial expression + vocal tone fusion |
| Processing Requirements | High-end processors, industrial-grade cameras/mics | High-end processors, dedicated GPU | Moderate to high | Lightweight, runs on affordable embedded processors |
| Real-Time Performance | Yes, but can degrade in noisy/light-variable conditions | Yes, but with occasional latency in emotion updates | Often near-real time in labs, but less tested in homes | Optimized for low-latency real-time response in dynamic home environment |
| Noise/Lightning Robustness | Moderate | Moderate | Limited in many studies | Pre-processing for noise suppression and illumination normalization |
| Empathetic Response Style | Scripted, predefined phrases | Scripted with some adaptive prompts | Usually scripted | Dynamic, context-aware empathetic speech generation |
| Cost & Accessibility | High | High | Prototype-only | Low-cost, accessible for personal home use |
| Assistive Care Focus | General-purpose, social robot | Elderly care with cognitive support | Usually proof-of-concept research | Specifically designed for assistive home care with simple, natural interaction |

## 1.4 RESEARCH PROBLEM

Most emotion-aware social robots currently on the market or in research. Such as Pepper, Ryan, and various academic prototypes are designed for broad, multi-purpose interaction in relatively controlled environments. While these systems can detect a wide range of emotions and engage in social conversation, they often face limitations when applied to assistive home environments that are dynamic, unpredictable, and resource-constrained.

A major challenge lies in emotion detection accuracy and responsiveness. Many systems attempt to recognize six or more emotions, which increases computational load and reduces reliability in low-light or noisy conditions. This is problematic in real homes, where ambient noise, changing illumination, and partial occlusions are common. Delayed or incorrect emotion recognition in such contexts can result in irrelevant or even emotionally insensitive responses, undermining user trust.

In addition, existing response-generation mechanisms are often scripted and lack true adaptability to the detected emotional state. They may repeat predefined phrases that feel artificial or disconnected from the user's current context, reducing the sense of empathy that is essential in assistive care.

Some of the key unresolved challenges in this domain include:

- **Focused emotion detection –** Optimizing for only the most relevant emotional states in assistive care to improve speed and reliability.
- **Robustness to home conditions –** Maintaining accuracy under varying lighting, noise, and partial occlusion.
- **Adaptive empathetic response –** Generating dynamic speech tailored to the detected emotion, rather than relying solely on fixed scripts.
- **Low-cost accessibility –** Enabling deployment on affordable, resource-limited hardware without sacrificing real-time performance.

This research seeks to address these limitations by developing a lightweight, real-time emotion-aware interaction system capable of detecting happiness and sadness through fusion of facial expression and vocal tone analysis. The system will dynamically generate empathetic speech responses tailored to the detected emotion and will be designed for deployment in real-world assistive home environments, balancing robustness, accessibility, and emotional relevance.

# 2. RESEARCH OBJECTIVES

## 2.1 MAIN OBJECTIVE

The primary objective of this research is to design, implement, and evaluate a real-time, emotion-aware multimodal interaction system for assistive robots, capable of detecting happy and sad emotions from facial expressions, and voice tone, and responding with empathetic, context-appropriate speech. The system will be optimized for low-cost hardware and assistive home environments, enabling natural, accessible, and emotionally intelligent interaction for elderly individuals, people with disabilities, and non-technical users.

## 2.2 SPECIFIC OBJECTIVES

**Develop a Real-Time Emotion Detection Framework**

- Implement a lightweight computer vision pipeline using OpenCV and a CNN-based facial expression recognition model, optimized for two-emotion classification.
- Integrate speech signal processing and prosody analysis to detect emotional cues from voice tone in real time.
- Apply feature fusion techniques to combine facial and vocal emotion data for higher detection accuracy.

**Design a Multimodal Command Fusion Engine**

- Create a decision-making engine to merge voice, and emotion inputs into unified, context-aware control signal.
- Implement conflict resolution rules to ensure safety
- Allow emotion detection to dynamically influence robot behavior and speech tone

**Implement an Empathetic Response Generator**

- Use text-to-speech (TTS) to deliver emotion-adaptive spoken responses that vary in tone, pitch, and word choice.
- Develop context-aware dialogue strategies for responding to happy or sad states in a supportive and non-repetitive manner.

**Optimize for Low-Cost, Real-Time Deployment**

- Ensure all modules run efficiently on resource-limited hardware
- Minimize processing latency to enable real-time human-robot interaction without noticeable delay

# 3. METHODOLOGY

This research follows a modular, iterative methodology to design, implement, and evaluate a real-time multimodal emotion-aware interaction framework for assistive robots. The process integrates facial expression recognition and vocal tone analysis into a unified decision-making system that enables empathetic, adaptive responses in dynamic home environments.

## 3.1 Multimodal Emotion and Command Detection

### 3.1.1 Facial Emotion Recognition (FER)

- **Objective:** Detect and classify user emotions into happy or sad in real time
- **Approach:**
    - Capture facial video using an RGB camera mounted on the robot
    - Preprocess frames with OpenCV for face detection and cropping.
    - Extract facial features using a lightweight CNN or MobileNetV2 model trained on FER2013 or CK+ datasets, fine-tuned for the two-class emotion subset

### 3.1.2 Vocal Tone Analysis

- **Objective:** Extract emotional state and intent from voice tone and speech content.
- **Approach:**
    - Use a microphone array to capture audio while minimizing background noise via spectral subtraction or RNNoise filtering.
    - Apply a speech-to-text (STT) model (Vosk or DeepSpeech) for command transcription
    - Parallelly, extract prosodic features (pitch, energy, MFCCs) for tone-based emotion classification using an SVM or lightweight neural classifier.

## 3.2 Multimodal Fusion and Context-Aware Decision Making

- **Objective:** Merge data from all input channels into a single, coherent decision that accounts for both commands and emotional context.
- **Approach:**
    - Introduce Emotion-Weighted Command Modulation, where detected emotion influences how the robot responds:
        - If sad: softer, slower voice, more empathetic responses.
        - If happy: more energetic, conversational tone.

## 3.3 Empathetic Speech Response Generation

- **Objective:** Generate verbal feedback that is both contextually correct and emotionally tuned.
- **Approach:**
    - Use a text-to-speech (TTS) engine with adjustable pitch, pace, and tone to reflect emotional alignment.

## 3.4 System Architecture Overview

The system follows a ROS2-based modular architecture:

1. Perception Layer – Camera and microphone nodes, pre-processing filters, and feature extraction modules for FER, and voice.
2. Fusion & Decision Layer – Multimodal fusion node, emotion-weighted decision engine, and safety/conflict resolution.
3. Interaction Layer – TTS node for speech output, behavior execution module, and user feedback loop
4. Integration Layer – ROS2 topics and services ensuring low-latency communication between modules

## 3.5 Project Flow

1. Data capture – Camera and mic continuously collect input.
2. Preprocessing – Noise removal, image normalization.
3. Individual Recognition – FER, vocal tone analysis.
4. Multimodal Fusion – Inputs combined into unified emotional-command vector.
5. Decision making – Context-aware selection of actions, prioritizing safety.
6. Response execution – Robot moves, speaks, or changes behavior accordingly.

## 3.6 Evaluation Plan

- **Accuracy Metrics:**
    - Emotion classification accuracy (FER and voice tone)
- **Performance Metrics:**
    - End-to-end latency from input capture to response (<500ms target)
- **User Study:**
    - Recruit participants from diverse age groups to rate interaction naturalness, trust, and emotional appropriateness.

## 3.7 Commercialization

The A.L.T.R.U.S care robot will be promoted as an end-to-end, modular ROS2-based solution specifically designed for elderly care homes, smart homes, hospitals, and rehabilitation centers. Its new blend of cutting-edge features—Active SLAM with reinforcement learning adaptive wayfinding, real-time health and environmental monitoring with multi-sensor fusion, multi-modal affect detection and empathetic behavior synthesis, and fault-tolerant secure middleware with blockchain-based logging—makes it highly adaptable to various care environments. The platform will be marketed both as a complete standalone robot and as upgrade packs (voice & gesture control pack, monitoring module, emotion sensing kit) for existing domestic or service robots in order to enable cost-effective adoption. Commercialization will be targeted at strategic partnerships with healthcare organizations, elderly care facilities, and smart home integrators and also research validation partnerships with universities. Real-world pilot demonstrations will generate credibility, and targeted marketing campaigns, conference exhibits, and industry shows will make A.L.T.R.U.S a credible, engaging, and future-oriented assistive robotics solution that meets the expanding demand for safe, autonomous, and user-centric caregiving technology.

## 3.8 Overall System Diagram

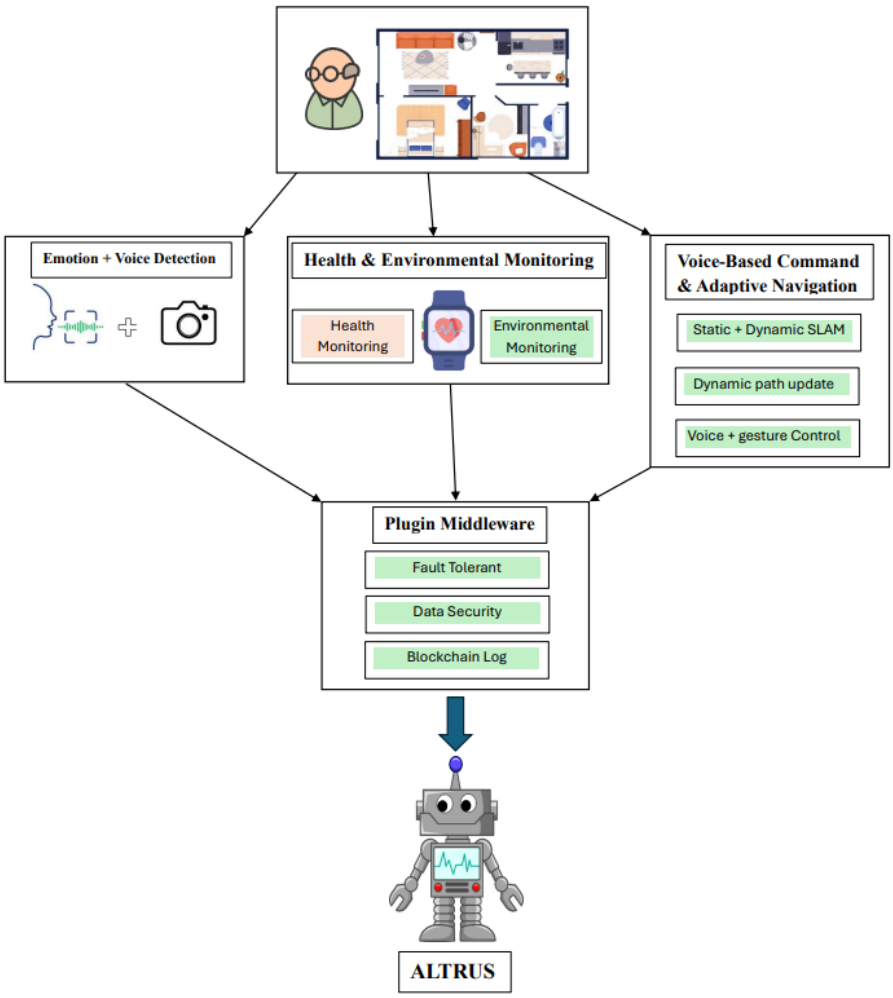



*Figure 4.3.1 System Overview Diagram*

# 4. PROJECT REQUIREMENTS

## 4.1 Functional Requirements

**Emotion Detection from Facial Expressions**

- The system shall detect happy and sad emotions from live camera input using a facial expression recognition model.
- The system shall work in real time with minimal latency.

**Emotion Detection from Voice Tone**

- The system shall process live microphone input to analyze vocal tone, pitch, and energy levels.
- The system shall classify emotions into happy or sad from voice alone when facial input is not available

**Multimodal Emotion Fusion**

- The system shall combine facial expression and vocal tone data to product a unified emotion classification output.
- The fusion process shall prioritize the more reliable modality in cases of conflicting signals.

**Multimodal Command Fusion**

- The system shall merge voice commands and detected emotions into a single context-aware decision.
- The system shall apply safety rules.

**Empathetic Response Generation**

- The system shall generate spoken responses using TTS, adjusting tone and vocabulary according to detected emotion
- The system shall deliver responses within 1.5 seconds of emotion detection.

**Real-Time Interaction**

- The system shall operate continuously in real time, processing all inputs and outputs without noticeable delays.

**User Accessibility**

- The system shall allow operation without requiring mobile apps or physical controllers, relying on natural inputs.

## 4.2 Non-Functional Requirements

**Performance**

- Emotion recognition accuracy shall be >= 85% under normal lightning and noise conditions.
- Average system response time shall be <= 1.5 seconds.

**Reliability**

- The system shall maintain stable operation for at lease 8 continuous hours without performance degradation.
- The system shall handle temporary signal loss without crashing.

**Scalability**

- The system shall allow adding more emotions without major changes to the architecture

**Portability**

- The system shall run on low-cost embedded hardware (e.g., Raspberry Pi 5)
- The system shall be deployable across different ROS2-compatible robots

**Usability**

- Users with no technical background shall be able to operate the system after a <= 5-minute introduction.
- The user interface shall use clear, simple language.

**Safety**

- The system shall stop all motion immediately upon detecting a stop gesture or urgent voice command.
- No physical movement shall occur if the emotion fusion engine detects user distress.

**Robustness**

- The system shall operate under variable indoor lightning.
- The system shall maintain >=80% accuracy in the presence of moderate background noise.

# 5. BUDGET DETAILS

| Item | Estimated Cost (LKR) |
|---|---|
| Raspberry Pi 5 8GB Original UK | Rs.39,950.00 |
| RPLiDAR 2D | Rs.37,900.00 |
| PK-910H 1080p Full-HD WebCam | Rs.8,000.00 |
| Wheels | Rs.5,000.00 |
| Server motor 2 (Mg90s) | Rs.1,600.00 |
| Gear motor 2 (L298) | Rs.6,000.00 |
| Motor controller | Rs.500.00 |
| ESB 32 | Rs.1,700.00 |
| Oled display | Rs.600.00 |
| Heart rate + blood oxygen | Rs.600.00 |
| Ecg | Rs.1,600.00 |
| Accelerometer + gyroscope | Rs.1,500.00 |
| body temperature | Rs.2,500.00 |
| lipo battery | Rs.2,500.00 |
| Gas sensor + Fire sensor | Rs.1,000.00 |
| Additional Cost | Rs.30,000 |
| Total Cost | Rs.140,950.00 |

*Table 1 budget for the proposed system*

# 6. CONCLUSION

The proposed Real-Time Emotion-Aware Multimodal Interaction Model for Assistive Robots addresses a critical gap in current human-robot interaction systems by integrating facial expression recognition, vocal tone analysis into a unified, real-time decision-making framework. Unlike most existing assistive robots, which rely on single-modal inputs or treat voice and gesture control as independent features, this system fuses emotional and command data to enable context-aware, empathetic responses.

By focusing on two primary emotional states (happy and sad) in the system ensures practical feasibility while retaining strong real-world relevance, particularly for elderly individuals, people with disabilities, and non-technical users. The design prioritizes natural, intuitive interaction, eliminating the need for mobile apps or physical controllers, and emphasizes fast response times, robust performance under variable lighting and noise conditions, and safety in dynamic environments.

This research contributes novelty by coupling multimodal emotion detection with an empathetic speech response system, creating a more human-like interaction experience for assistive robots. The result is a more trustworthy, responsive, and emotionally intelligent robotic assistant that can adapt to user states in real time.

Through implementation in an ROS2-compatible framework, the system offers portability and scalability, making it adaptable for both academic research platforms and commercial assistive robots. Ultimately, this work represents a meaningful step toward closing the gap between purely functional robotics and emotionally aware companions capable of enriching daily human life.

# REFERENCES

## References


[1] Y. e. a. Li, "Lightweight Convolutional Neural Network for Real-Time Facial Expression Recognition," *Sensors,* vol. 22, no. 9, p. 3336, 2022.

[2] B. e. a. Jiang, "Real-Time Facial Expression Recognition for Human–Robot Interaction Using Residual Networks," *Applied Sciences,* vol. 13, no. 4, p. 2156, 2023.

[3] Y. e. a. Sari, "Facial Emotion Recognition in Social Robots for Autism Therapy," *Multimedia Tools and Applications,* 2023.

[4] P. e. a. Tzirakis, "End-to-End Speech Emotion Recognition Using Deep Neural Networks," *IEEE Journal of Selected Topics in Signal Processing,* vol. 14, no. 2, pp. 284-294, 2021.

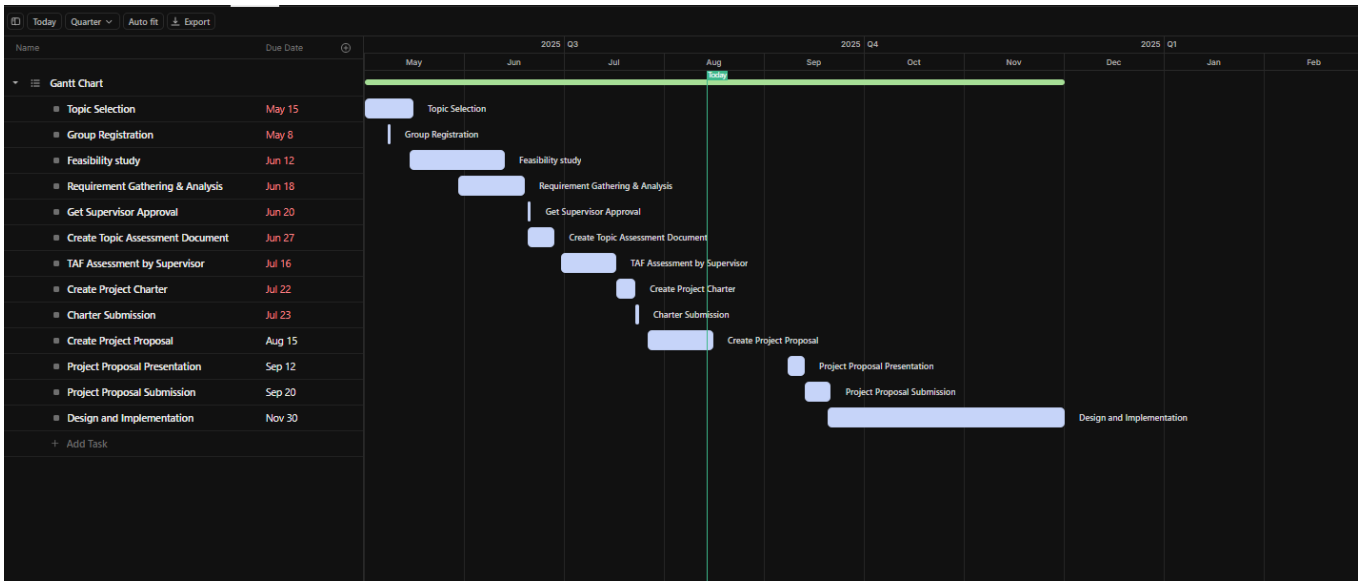
[5] H. &. Z. X. Li, "Context-Aware Speech Emotion Recognition Using NLP and Acoustic Features," *Speech Communication,* vol. 150, pp. 101-115, 2024.

[6] S. e. a. Anwar, "Multimodal Emotion Recognition Using Facial and Speech Cues for Human–Robot Interaction," *IEEE Access,* vol. 11, pp. 28456-28469, 2023.

[7] L. e. a. Chen, "Transformer-Based Multimodal Fusion for Emotion Recognition in Human–Robot Interaction," *Sensors,* vol. 24, no. 2, p. 512, 2024.

# APPENDICES

## Appendix 1. Gantt Chart



# Appendix 2. Online Survey

## Welcome to the A.L.T.R.U.S Assistive Robot Survey

Thank you for taking the time to participate in our research survey!

**About Our Research**

Our project is titled **"A.L.T.R.U.S" – Adaptive Life-support & Therapeutic Robotic Unit System**. We are developing an advanced assistive robot designed to support home healthcare by providing safe, intelligent, and empathetic assistance to users with medical or mobility needs.

The goal of A.L.T.R.U.S is to create a robust robot that can:

- Understand and respond to voice and gesture commands naturally
- Monitor health and environmental conditions in real time
- Recognize user emotions and respond empathetically
- Recover from faults and update itself securely

**The Four Core Components of A.L.T.R.U.S**

**Voice-Based Command Interpretation & Adaptive Navigation :** The robot understands voice commands and uses smart navigation to move safely around obstacles and adapt to changes in the home environment.
**Health and Environmental Monitoring Using Sensor Fusion :** Multiple sensors work together to continuously monitor vital signs (like heart rate and temperature) and environmental factors (like gas leaks or air quality) for early emergency detection.
**Emotion Detection & Empathetic Interaction :** The robot recognizes emotions from facial expressions and voice tone, combining these inputs to respond with empathetic speech and appropriate gestures.
**Fault-Tolerant Middleware, Secure Updates & Blockchain Logging :** A fault-tolerant system ensures continuous operation even during errors, securely manages software updates, and maintains tamper-proof logs of the robot's activities.

Your feedback is crucial to help us understand which features matter most to potential users. The survey will take about 10 minutes to complete.

Thank you for contributing to this important research!

Experience with Robots or Smart Devices: *

- ○ None
- ○ Basic (e.g., using smart assistants like Alexa/Siri)
- ○ Moderate (e.g., programming robots, using advanced devices)
- ○ High (e.g., robotics development or engineering)

How important is it for the robot to move safely even if the room layout changes (e.g., furniture moved)? *

- ○ Extremely important – safety comes first
- ○ Important – smooth movement matters
- ○ Somewhat important – small mistakes are okay
- ○ Not important – I can guide it myself

What would worry you most about using a robot that moves on its own?" *

- ☐ Bumping into people or objects
- ☐ Not understanding commands
- ☐ Moving to the wrong place
- ☐ Being too slow to react
- ☐ Needing frequent recharging

If the robot could understand both your voice and gestures, would that make it easier for you to use? *

- ○ Yes, much easier
- ○ Yes, a little easier
- ○ No difference
- ○ No, I prefer only one control method

Which features would you like most in a home robot with voice and gesture control? *

- ☐ Understand my voice commands accurately
- ☐ Recognize my hand gestures quickly
- ☐ Avoid obstacles without my help
- ☐ Adapt if the furniture is moved
- ☐ Come to me when I call it

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

Short-answer text

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

- Very familiar, I use one regularly
- Somewhat familiar, I've heard about them but never used one
- Not familiar at all

If a wristband alerted you about a possible medical emergency, how likely are you to act immediately (e.g., call emergency services, inform family)? *

- Very likely
- Likely
- Neutral
- Unlikely
- Very unlikely

Which of these features would you find most useful in a health emergency detection wristband? *

- [ ] Heart attack detection
- [ ] Tachycardia & bradycardia detection
- [ ] Cardiac arrest detection
- [ ] Fall detection
- [ ] Fever detection
- [ ] Emergency call/SMS alerts
- [ ] Historical health trend analysis

What concerns would you have about using a multi-emergency detection wristband?" *

- [ ] Accuracy of health data
- [ ] Privacy of personal health information
- [ ] Battery life
- [ ] Comfort and wearability
- [ ] Cost of the device
- [ ] Risk of false alarms

Have you ever interacted with a robot or virtual assistant? *

- Yes
- No

How would you rate your comfort level interacting with robots? *

- Very comfortable
- Somewhat comfortable
- Neutral
- Uncomfortable
- Very uncomfortable

Which emotions do you think are most important for a robot to detect? *

- [ ] Happiness
- [ ] Sadness
- [ ] Anger
- [ ] Fear
- [ ] Surprise
- [ ] Disgust

Which emotional cues do you think a robot should detect? *

- [ ] Facial expression
- [ ] Tone of voice
- [ ] Gestures/body language
- [ ] Words used in speech

Would you trust a robot that can read your emotions? *

- ( ) Yes
- ( ) Maybe
- ( ) No

If a robot was helping you at home, what tasks would you want it to do without ever failing? *

Short-answer text

How important is it for an assistive robot to keep working even if some parts stop working? *

- ( ) Extremely important – safety comes first
- ( ) Important – reliability matters
- ( ) Somewhat important – minor issues are okay
- ( ) Not important – I can manage without it

If a robot kept a secure, unchangeable record of what it did, would this make you trust it more? *

- ( ) Yes, a lot more
- ( ) Yes, a little more
- ( ) No difference
- ( ) No, it wouldn't matter

Which of these would you want in a helpful robot? *

- [ ] Understand my requests correctly
- [ ] Fix itself when something goes wrong
- [ ] Keep a secure record of actions
- [ ] Connect different robot features smoothly (e.g., movement + health checks)
- [ ] Share information instantly between parts

What would worry you about using a robot that makes its own decisions? *

- [ ] Safety during malfunctions
- [ ] Misunderstanding commands
- [ ] Privacy of personal information
- [ ] Relying too much on technology
- [ ] Cost of buying or maintaining it
- [ ] Being too complicated to use

From a robotics systems perspective, what core services should middleware guarantee in a mission-critical assistive robot? *

Short-answer text

In modular assistive robots, where would you rank fault tolerance in the middleware priority list? *

- ( ) Top priority – essential for safe autonomy
- ( ) High priority – crucial for user trust
- ( ) Medium priority – nice to have but not critical
- ( ) Low priority – can be handled at application level

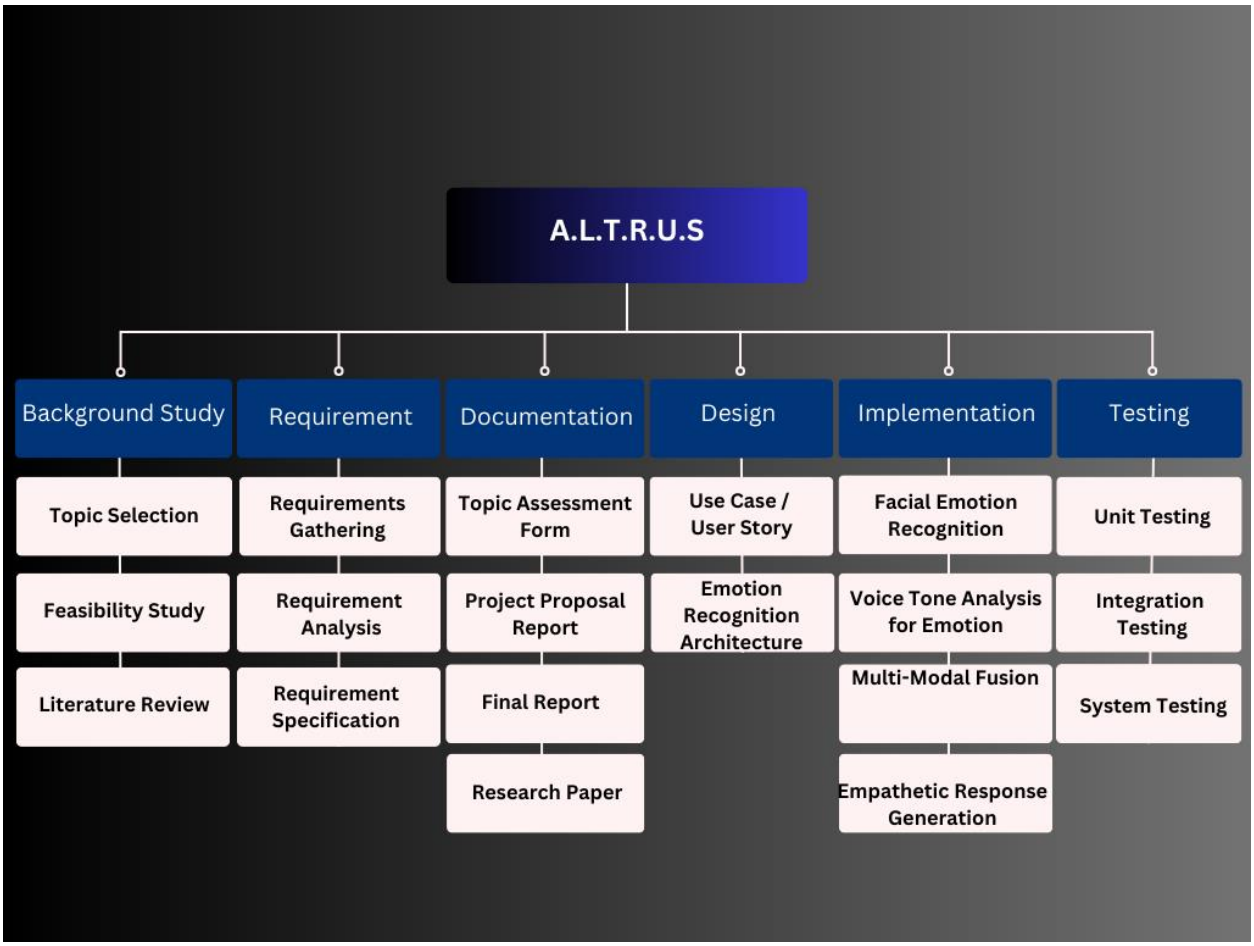
How effective do you think blockchain is for ensuring data integrity and accountability in distributed robotic systems? *

- ( ) Very effective – ideal for immutable logging
- ( ) Effective – works well with limitations
- ( ) Neutral – other methods are equally good
- ( ) Ineffective – adds unnecessary complexity

Which features would you expect from a next-generation robot middleware? *

- [ ] Intent recognition and context mapping
- [ ] Real-time fault detection and self-healing
- [ ] Blockchain-based action/event logging
- [ ] Modular plugin management for robot capabilities
- [ ] Developer tools for monitoring and debugging

What challenges do you foresee in implementing fault-tolerant, blockchain-backed middleware for modular robots? *

- [ ] Real-time performance overhead
- [ ] High resource consumption (CPU, storage)
- [ ] Complexity of fault recovery algorithms
- [ ] Blockchain scalability and latency
- [ ] Interoperability between heterogeneous modules
- [ ] Developer adoption and learning curve

## Appendix 3. Work Breakdown Structure